

ICSA 2026 SHORT COURSE

## Lecture 3: Covariate shift and density-ratio reweighting

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*Statistical Foundations of Transfer Learning*

YANG FENG<sup>1</sup>   YE TIAN<sup>2</sup>

<sup>1</sup>NYU

<sup>2</sup>Yale University

# Overview

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§3.1: Covariate shift

§3.2: Is covariate shift adaptable for free?

§3.3: The reweighting method

Numerical illustration: importance weighting

§3.4: Density-ratio estimation

References

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## §3.1: Covariate shift

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## Covariate shift, formally

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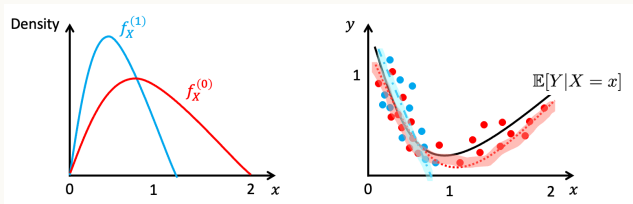
*In  $L2$  the covariate-shift gap was a fixed penalty  $2d_{TV}$  we could only bound; this lecture shows importance weighting removes it and recovers oracle rates.*

- Source  $(X, Y) \sim \mathbb{P}^{(1)}$ , target  $(X, Y) \sim \mathbb{P}^{(0)}$ , on  $\mathcal{X} \times \mathcal{Y}$ .

- **Covariate shift:**

$$\mathbb{P}_X^{(0)} \neq \mathbb{P}_X^{(1)}, \quad \mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)} \quad (\text{a.s. } \mathbb{P}_X^{(0)}).$$

- Equivalent: there exists a regression / decision function  $f^*$  shared by source and target; only the feature distribution changes.



## Three motivating scenarios

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- **Self-driving cars:** training routes  $\subsetneq$  deployment routes. Same physics; new lighting, weather, traffic distributions.
- **Medical imaging:** Alzheimer's from MRI: a research-study scanner (source) vs. a community-hospital scanner (target) — same disease, shifted images (our L1 running example).
- **A/B testing for ML:** the production traffic mix shifts after a model is shipped, but the conditional model of  $Y \mid X$  is unchanged.

**Why special:** in all three, the *physics/biology/intent* is fixed — only the *distribution of situations* changes. That is covariate shift.

## Density ratio: the central object

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Define the source-to-target density ratio

$$w(\mathbf{x}) = \frac{d\mathbb{P}_X^{(0)}}{d\mathbb{P}_X^{(1)}}(\mathbf{x}),$$

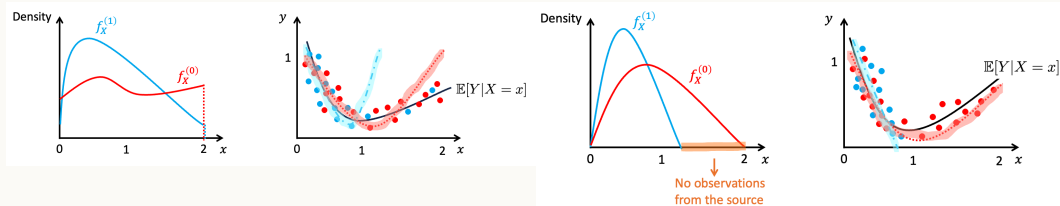
assuming  $\mathbb{P}_X^{(0)} \ll \mathbb{P}_X^{(1)}$  (target absolutely continuous w.r.t. source).

- Intuition:  $w(\mathbf{x}) > 1$  at points the target “cares about” more than the source.
- Failure case:  $\sup_{\mathbf{x}} w(\mathbf{x}) = \infty$ . Then any source-only estimator can be fooled.
- Key fact: for any integrable  $g$ ,  $\mathbb{E}_{\mathbb{P}^{(0)}}[g(X)] = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)g(X)]$ .
- *Think of it as:* reweighting source points by  $w$  lets you “pretend” you sampled from the target — importance sampling over  $\mathcal{X}$ .

# Bounded vs. unbounded density ratios

Bounded  $w$  (well-behaved)

Unbounded  $w$  (failure)



Bounded  $w$  + small effective sample size  $n_1/\|w\|_\infty \Rightarrow$  usable transfer.

*Unbounded  $w$ : a single source point in a region the source rarely visits but the target loves gets an astronomically large weight — one observation dominates the empirical risk.*

## Check-in: what defines covariate shift?

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*In which of the following scenarios is the assumption  $\mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)}$  most plausible?*

- (a) Survey of US households 2010 vs. 2020:  $X$  is income,  $Y$  is voting preference.
- (b) Camera sees same kind of objects under new lighting;  $X$  is the pixel array,  $Y$  is the object class.
- (c) Hospital A's MRI vs. Hospital B's CT scan;  $Y$  is whether the patient has tumor.
- (d) Pre- vs. post-pandemic search queries;  $Y$  is whether to click a particular ad.

### Answer

(b). Same physical objects under different lighting – the underlying object-given-pixel mapping is unchanged. (a) and (d) involve a shifting world (people change their preferences), and (c) doesn't even share the feature space ( $X$  is different).



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§3.2: Is covariate shift adaptable for free?

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## When ERM survives covariate shift unaided

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Consider parametric MLE under a **well-specified** model:  $\mathbb{P}_{Y|X}^{(k)} = p_\theta(Y|X)$  with the same  $\theta$  for  $k = 0, 1$ .

- The MLE on source converges to  $\theta^*$  regardless of  $\mathbb{P}_X^{(1)}$  (as long as it puts mass on the relevant directions).
- Plug in  $\hat{\theta}$  to get  $\hat{p}_{Y|X}$ ; this works on the target by assumption.
- Conclusion: **well-specified parametric MLE is automatically robust to covariate shift.**
- *Why?* MLE maximizes  $\sum_i \log p_\theta(Y_i | X_i)$  — a conditional loss; the score  $\nabla_\theta \log p_\theta(Y | X)$  depends only on  $P_{Y|X}$ , so the source marginal is irrelevant.
- In fact source-only MLE is *minimax-optimal* on the target under a well-specified model — with *no* bounded-density-ratio assumption (Ge et al., 2024).

## When ERM fails: mis-specification or non-parametric

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Two regimes where naive ERM is biased on the target:

- **Mis-specified model:** the best linear (etc.) approximation is determined by  $\mathbb{P}_X^{(k)}$ .  
Different  $\mathbb{P}_X^{(k)} \Rightarrow$  different “best” parameter  $\Rightarrow$  different prediction.
- **Non-parametric estimators** (kernel, nearest neighbor): convergence rates depend on the local density of training points. Source-skewed densities give source-skewed accuracy.

**Takeaway:** “free” covariate-shift adaptation is the exception, not the rule. When in doubt, correct.

## Beyond parametric models: $k$ -NN under covariate shift

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Suppose  $f^*$  is  $\beta$ -Hölder and  $\mathbb{P}_X^{(1)}$  has support strictly containing  $\mathbb{P}_X^{(0)}$ . Then  $k$ -NN regression has target risk

$$R^{(0)}(\hat{h}_{kNN}) - R^{(0)}(f^*) \lesssim \mathbb{E}_{X \sim \mathbb{P}^{(0)}} [w(X)^\beta] \cdot n_1^{-2\beta/(2\beta+d)}$$

where  $w(X) = \frac{d\mathbb{P}_X^{(0)}}{d\mathbb{P}_X^{(1)}}(X)$  is the density ratio (recall  $w(X) = d\mathbb{P}_X^{(0)}/d\mathbb{P}_X^{(1)}$ , the same object reweighting uses) — large exactly where the source is sparse, so the constant  $\mathbb{E}[w^\beta]$  (not the rate) is what covariate shift inflates.

*Why  $w^\beta$ ? A  $\beta$ -Hölder  $f^*$  makes a neighbor at distance  $r$  contribute bias  $\sim r^\beta$ ; where source is sparse ( $w$  large) the  $k$ -NN radius is large, so bias  $\sim w^\beta$  — hence  $\mathbb{E}[w^\beta]$ .*

- If  $\mathbb{P}_X^{(0)}$  concentrates where source is sparse,  $\mathbb{E}[w^\beta]$  blows up.
- This is the non-parametric face of covariate shift: the rate is right, the constant explodes.

**Fix:** reweight source points so they mimic the target density — the reweighting method, next.

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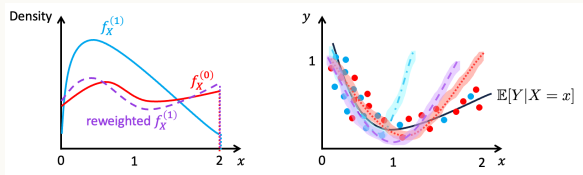
## §3.3: The reweighting method

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# Reweighted ERM

Fix a loss  $\ell(h(X), Y)$  and recall  $\mathbb{E}_{\mathbb{P}^{(0)}}[\ell] = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell]$ . Define the **importance-weighted ERM**:

$$\hat{h}_{\text{IW}} = \arg \min_{h \in \mathcal{H}} \frac{1}{n_1} \sum_{i=1}^{n_1} w(\mathbf{x}_i^{(1)}) \ell(h(\mathbf{x}_i^{(1)}), y_i^{(1)}).$$



Source points reweighted by  $w(\mathbf{x}_i)$  (larger = “more like target”); after reweighting the empirical law mimics the target.

## Why reweighting works

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### Theorem 3.3.1 (Cortes et al., 2010)

*The IW objective is unbiased for the target risk, and the deviation is governed by the second-moment / Rényi divergence  $d_2(\mathbb{P}^{(0)} \parallel \mathbb{P}^{(1)}) = \mathbb{E}_{\mathbb{P}^{(1)}}[w(X)^2]$ : w.p.  $1 - \delta$ ,*

$$R^{(0)}(\hat{h}_{\text{IW}}) \leq \min_h R^{(0)}(h) + \mathcal{O}\left(\sqrt{\frac{d_2(\mathbb{P}^{(0)} \parallel \mathbb{P}^{(1)}) \text{VC}(\mathcal{H})}{n_1}}\right).$$

- Unbiased:  $\mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell(h(X), Y)] = R^{(0)}(h)$ .
- *Why?* Under covariate shift  $\mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)}$ , so
$$\mathbb{E}_{\mathbb{P}^{(1)}}[w(X)\ell] = \int w d\mathbb{P}_X^{(1)} \int \ell d\mathbb{P}_{Y|X}^{(1)} = \int d\mathbb{P}_X^{(0)} \int \ell d\mathbb{P}_{Y|X}^{(0)} = R^{(0)}(h).$$



## Why reweighting works (cont.)

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- “Effective sample size” =  $n_1/d_2$ : heavy-tailed  $w$  (large  $d_2$ ) wastes data. (Bounded  $w \leq W$  is the special case  $d_2 \leq W$ .)
- *Why  $d_2$ ?* The IW risk has variance  $\propto \mathbb{E}_{\mathbb{P}^{(1)}}[w^2 \ell^2]/n_1 \leq \mathbb{E}_{\mathbb{P}^{(1)}}[w^2]/n_1 = d_2/n_1$  — noise scales with the *second moment* of the weight, not its max.
- Unbounded  $w$ : the rate degrades below  $\sqrt{n_1}$ ; in practice  $w$  is unknown  $\Rightarrow$  density-ratio estimation (§3.4).

## Check-in: importance weighting

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*The true density ratio is  $w(\mathbf{x}) = e^{2\mathbf{x}_1}$ , so  $\sup w = \infty$  on  $\mathbb{R}^d$ . Your source has  $n_1 = 10,000$  points.*

Which fix is principled?

- (a) Use IW-ERM; the unbiased estimator ensures consistency.
- (b) Clip  $w$  at some  $W$ , accept a small bias, recover variance control.
- (c) Switch to target-only ERM; we have no labels, so we are stuck.
- (d) Use both methods and take the minimum-MSE estimator.

### Answer

(b). Unbounded  $w$  blows up the IW variance: a single source point with very large weight can dominate the empirical risk. Truncating at some level  $W$  yields a tractable bias-variance trade-off (also justified theoretically when  $w$  has heavy tails).

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**Numerical illustration: importance weighting**

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## Numerical illustration: setup

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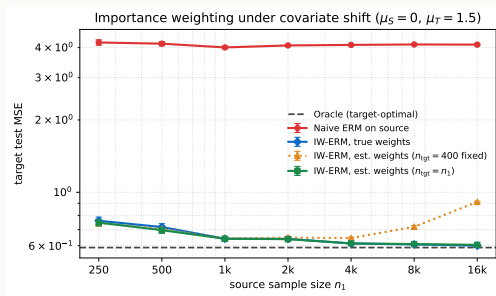
Source  $X \sim N(0, 1)$ , target  $X \sim N(1.5, 1)$  (so  $w(x) = e^{1.5x - 1.125}$  [ratio of  $N(1.5, 1)$  to  $N(0, 1)$  densities, so  $w(x) = \exp(1.5x - 1.5^2/2)$ ], bounded on bounded sets). Truth  $y = 0.5x^2 + 0.3\varepsilon$ , fit class is *linear* (mis-specified).

- Source OLS:  $b \approx 0$  (since  $\text{Cov}(x, x^2) = 0$  at  $\mu_S=0$ ).
- Target OLS:  $b \approx 1.5$  (best linear fit around  $\mu_T=1.5$ ).

Compare on the target test set against the **oracle** (population target-optimal fit): naive ERM; IW-ERM with *true* weights; and IW-ERM with *estimated* weights (logistic discriminator), using either a *fixed* unlabeled-target pool ( $n_{\text{tgt}}=400$ ) or one that *grows* with  $n_1$ .

Code: `sims/L3_importance_weighting.py`.

## Numerical illustration: result



- Naive ERM is biased by  $\sim 4\times$  the irreducible target MSE — more source never helps.
- IW-ERM with the *true* weights recovers the oracle as  $n_1$  grows.
- Estimated weights from a *fixed* target pool ( $n_{tgt}=400$ ) plateau *above* the oracle — the weights are only as good as that target sample.
- Grow the target with  $n_1$  and that curve drops to the oracle, matching true weights.

## Numerical takeaway

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- Under covariate shift + mis-specification, naive ERM is *permanently biased*: more source data does not help.
- IW reweighting is the right correction; it converges to oracle at the standard  $\sqrt{n_1}$  rate, scaled by the Rényi divergence  $d_2(\mathbb{P}^{(0)} \parallel \mathbb{P}^{(1)}) = \mathbb{E}_{\mathbb{P}^{(1)}}[w^2]$ .
- Estimated weights from a binary discriminator work well *given enough unlabeled target* to learn them (previewed in §3.4, the discriminative method).

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## §3.4: Density-ratio estimation

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## Estimating the density ratio: four canonical routes

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*Everything so far assumed  $w$  known; now we estimate it from source plus unlabeled target features. The four routes are surrogates for the same  $w$ : match moments (KMM), classify domains (discriminative), or fit  $w$  directly (KL / least-squares).*

Given source  $\{\mathbf{x}_i^{(1)}\}_{i=1}^{n_1}$  and unlabeled target  $\{\mathbf{x}_j^{(0)}\}_{j=1}^{n_0}$ :

- **Kernel mean matching** (Huang et al., 2006): match RKHS mean embeddings.
- **Discriminative learning** (Bickel et al., 2009): train a classifier of source vs. target; convert probabilities to a ratio.
- **KL-divergence minimization** (Sugiyama et al., 2008): fit a parametric  $\hat{w}$  to maximize the KL surrogate.
- **Least-squares** (Kanamori et al., 2009; Nguyen et al., 2010): solve a convex regression for  $w$  directly.

*(here  $n_0$  counts unlabeled target points — covariate-shift transfer has no target labels; contrast L2/L4 where  $n_0$  was labeled target examples.)*

## Density-ratio estimation: scope of the four routes (cont.)

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*The four routes are the canonical, dimension-robust estimators; the rest of this section develops each in turn.*

- **Kernel mean matching** — match RKHS mean embeddings (moments).
- **Discriminative** — classify source vs. target, convert to a ratio.
- **KL / least-squares** — fit  $w$  directly via a divergence surrogate.

Other approaches (separate KDE, histograms, semi-parametric) are useful in 1-D / low-dim or under structural assumptions; we summarize them in one slide at the end.

## Kernel mean matching (KMM)

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Idea: choose source weights  $\hat{w}_i$  so that the *reweighted* source mean embedding matches the target mean embedding in an RKHS  $\mathcal{H}_K$ :

$$\min_{\hat{w}_{\geq 0}} \left\| \frac{1}{n_1} \sum_{i=1}^{n_1} \hat{w}_i K(\mathbf{x}_i^{(1)}, \cdot) - \frac{1}{n_0} \sum_{j=1}^{n_0} K(\mathbf{x}_j^{(0)}, \cdot) \right\|_{\mathcal{H}_K}^2 \quad \text{s.t.} \quad \frac{1}{n_1} \sum_i \hat{w}_i = 1, \hat{w}_i \leq B.$$

- Convex QP in  $\hat{w}$ ; no parametric model for  $w$ .
- Connects directly to MMD (Lecture 2): the objective *is* an empirical squared MMD.
- *Picture*: treat  $K(\mathbf{x}, \cdot)$  as a “fingerprint” of  $\mathbf{x}$ ; KMM finds weights so the weighted-average source fingerprint equals the target’s — they look identical to any  $f \in \mathcal{H}_K$ .

## KMM: theory

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KMM is an algorithm (Huang et al., 2006); its *consistency* is analyzed later:

### Theorem 3.5.1 (Gretton et al., 2009; Yu and Szepesvári, 2012)

*With a universal kernel (i.e.  $\mathcal{H}_K$  dense in  $C(\mathcal{X})$ ), so matching means forces the full distributions to agree) and bounded weights  $\hat{w} \leq B$ , the reweighted feature mean converges ( $\hat{w} \rightarrow w$  in MMD as  $n_1, n_0 \rightarrow \infty$ ) and the IW-ERM with  $\hat{w}$  is consistent.*

- Practical note: tune  $B$  on a held-out validation set; popular choices  $B \in [10, 100]$ .
- Strength: works well in moderate dimension; nonparametric.
- Weakness: doesn't extrapolate beyond the source support; sensitive to  $B$ .

## Discriminative density-ratio learning

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Stack source (label  $z = 0$ ) and target (label  $z = 1$ ) and train any binary classifier

$\hat{\eta}(\mathbf{x}) = \widehat{\Pr}(z = 1 \mid \mathbf{x})$ . By Bayes,  $\Pr(z=k \mid \mathbf{x}) \propto p_X^{(k)}(\mathbf{x}) \Pr(z=k)$ , so

$\frac{p_X^{(0)}}{p_X^{(1)}} = \frac{\Pr(z=0|\mathbf{x})}{\Pr(z=1|\mathbf{x})} \cdot \frac{\Pr(z=1)}{\Pr(z=0)}$  (source  $z=0$ , target  $z=1$ ). Then by Bayes' rule,

$$\frac{\mathbb{P}_X^{(0)}(\mathbf{x})}{\mathbb{P}_X^{(1)}(\mathbf{x})} = \frac{\Pr(z = 1 \mid \mathbf{x})}{\Pr(z = 0 \mid \mathbf{x})} \cdot \frac{\Pr(z = 0)}{\Pr(z = 1)} \approx \frac{\hat{\eta}(\mathbf{x})}{1 - \hat{\eta}(\mathbf{x})} \cdot \frac{n_1}{n_0}.$$

- Any well-calibrated classifier (logistic, GBM, deep net) gives an estimate.
- **Easy to deploy**; the workhorse method in modern practice.
- Extends to non-tabular data (images, text) because the discriminator can.

*This is L2's source-vs-target domain classifier, now repurposed to estimate the weight  $w$  rather than just measure distance.*

## Discriminative density-ratio: theory

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### Theorem 3.5.2 (Bickel et al., 2009, informal)

*Suppose the discriminator class is rich enough and trained with  $n_1 + n_0$  joint samples; then with high probability,*

$$\left(\mathbb{E}_{\mathbb{P}^{(1)}} [(\hat{w} - w)^2]\right)^{1/2} \leq C \sqrt{\text{VC}(\mathcal{F})/(n_1 + n_0)} + \text{approximation error of } \mathcal{F}.$$

- Same rate as plain classification on  $z \in \{0, 1\}$  – no curse of dimensionality if the discriminator class is well chosen.
- Couples the IW-ERM analysis to standard classification theory.

## KL-divergence minimization (KLIEP)

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Parametrize  $w_\theta(\mathbf{x}) \geq 0$  (e.g.  $w_\theta(\mathbf{x}) = \exp(\theta^\top \phi(\mathbf{x}))$ ). Maximize

$$\frac{1}{n_0} \sum_j \log w_\theta(\mathbf{x}_j^{(0)}) \quad \text{s.t.} \quad \frac{1}{n_1} \sum_i w_\theta(\mathbf{x}_i^{(1)}) = 1 \quad (\text{enforces } \mathbb{E}_{\mathbb{P}^{(1)}}[w] = 1, \text{ i.e. } \hat{p}_0 \text{ is a density}).$$

Equivalent to minimizing  $\text{KL}(\mathbb{P}_X^{(0)} \parallel \hat{w} \cdot \mathbb{P}_X^{(1)})$  over the parametric class.

*Sketch:*  $\text{KL}(p^{(0)} \parallel \hat{w} p^{(1)}) = \mathbb{E}_{p^{(0)}}[\log p^{(0)}] - \mathbb{E}_{p^{(0)}}[\log \hat{w}] - \mathbb{E}_{p^{(0)}}[\log p^{(1)}]$ ; only the middle term depends on  $\hat{w}$ , so minimizing KL = maximizing  $\frac{1}{n_0} \sum_j \log w_\theta(\mathbf{x}_j^{(0)})$  s.t. normalization.

- **KLIEP** (Sugiyama et al., 2008): closed-form for log-linear  $w_\theta$ , with cross-entropy-type objective.
- Theory: under standard regularity,  $\hat{w}_\theta$  converges at the parametric rate.
- Caveat: parametric form for  $w$  may be restrictive, but works well for moderate  $d$ .

## Least-squares ratio estimation

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Direct convex regression for  $w$  itself:

$$\min_{w \in \mathcal{F}} \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [w(X)^2] - \mathbb{E}_{\mathbb{P}^{(0)}} [w(X)] = \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [(w - w^*)^2] + \text{const.}$$

*Derivation:*  $\frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [w^2] - \mathbb{E}_{\mathbb{P}^{(0)}} [w] = \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [w^2] - \mathbb{E}_{\mathbb{P}^{(1)}} [w w^*] = \frac{1}{2} \mathbb{E}_{\mathbb{P}^{(1)}} [(w - w^*)^2] - \text{const}$ , using  $\mathbb{E}_{\mathbb{P}^{(0)}} [g] = \mathbb{E}_{\mathbb{P}^{(1)}} [w^* g]$ .

Estimable by sample averages:  $\frac{1}{2n_1} \sum_i w(\mathbf{x}_i^{(1)})^2 - \frac{1}{n_0} \sum_j w(\mathbf{x}_j^{(0)})$ .

### Theorem 3.5.3 (Kanamori et al., 2009; Nguyen et al., 2010)

*For  $\mathcal{F}$  a kernel class,  $\hat{w}$  converges to  $w^*$  in  $L^2(\mathbb{P}^{(1)})$  at the kernel-regression rate.*

- Convex; supports kernels and neural-net parameterizations alike.
- Often more robust than KL methods when target has heavier tails than source.



## Other approaches (one-slide summary)

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| Method                    | Idea                                                                | When to use                      |
|---------------------------|---------------------------------------------------------------------|----------------------------------|
| Separate KDE              | Estimate $\hat{p}_0, \hat{p}_1$ , take ratio.                       | Low- $d$ , smooth densities.     |
| Histogram-based           | Bin the space; ratio = bin frequency.                               | Univariate; quick & dirty.       |
| Semi-parametric (Qin '98) | Tilt: $w_\theta = \exp(\theta^\top \phi)$ via empirical likelihood. | Moderate $d$ + structural prior. |

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### Practical guidance.

- Tabular, moderate  $d$ : KMM or KLIEP.
- Image / text / large  $d$ : discriminative.
- Heavy-tailed target: least-squares.
- Low- $d$  + smooth densities: separate KDE is fine and exploits structure.

## Check-in: which method should you pick?

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*You have 10,000 source images and 1,000 unlabeled target images at  $224 \times 224 \times 3$  resolution. You need  $\hat{w}$  for IW-ERM on a downstream image classifier.*

Which method is the obvious first choice?

- (a) Separate KDE on  $224^2 \times 3$ -dimensional pixel space.
- (b) KMM with a Gaussian kernel.
- (c) Train a CNN to classify source vs. target; convert logits to a ratio.
- (d) Histogram on principal components.

### Answer

(c), discriminative density-ratio estimation with a deep net. Curse-of-dimensionality kills (a) and (d); KMM is feasible but tuning  $B$  and the Gaussian bandwidth in 150,000-D is impractical. A binary classifier directly inherits the deep-net's representation.

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## Lecture summary

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- Covariate shift = different  $\mathbb{P}_X^{()}$ , same  $\mathbb{P}_{Y|X}^{()}$ .
- “Free” adaptation only for well-specified parametric MLE; otherwise ERM is biased.
- Reweight by  $w = d\mathbb{P}_X^{(0)} / d\mathbb{P}_X^{(1)}$  to recover the target objective.
- Bounded  $w$  + uniform deviation  $\Rightarrow$  standard rates with effective sample size  $n_1/d_2$ .
- Density-ratio estimation: KMM, discriminative, KL, least-squares (and friends).

Next: Lecture 4 turns to *posterior drift* –  $\mathbb{P}_X^{()}$  is shared but  $\mathbb{P}_{Y|X}^{()}$  differs – and asks how to bias an estimator toward the source as a route to “safe transfer.”

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## References

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## References

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